Local integration of GitHub QR/acoustic update

Date: **27 October 2026, Asia/Kolkata**. Incoming GitHub main: `404eed84cbb9f5f2eb0bc73e7b66910fb49476f9`, merged through PR #6. Feature implementation: `9a3640e2bc25f07da9dd450711c933f7d18624e7`. The prior local main (`3a92b73`) was clean and two commits behind, so it was fast-forwarded without conflicts or history rewriting. The incoming change covered 49 files, including the migration, protocol, UI, native modules and docs.

Updated runtime

- Installed the dashboard's locked dependencies, including QR rendering support.

- Created and checked a private PostgreSQL backup in the ignored work directory

before migrating. The backup contains private application data and is not an artifact.

- Upgraded the local database from `7a2d13ce4f90` to `9bd32a88f101`; Alembic model

drift check passed. Existing counts remained 4 users, 2 devices, 2 biometric profiles, 4 sessions, 4 attempts, 2 official records, 1 correction and 2 disputes.

- C003 coordinates/references and existing attendance outcomes were preserved.

iOS and unavailable-Wi-Fi policies remain review. No classroom challenge was automatically enabled and no new physical submission was made.

- Restarted the API and dashboard. API health/readiness passed, models loaded,

and the new `/api/v1/presence/evidence` route appeared in the running schema.

- Regenerated native projects with microphone permission/module declarations,

preserving local native settings. CocoaPods now includes SpvsAcoustic; the resolved Podfile.lock is retained in Git.

Local validation

Check	Local result
Backend	146 passed
Mobile	64 passed; TypeScript passed
Dashboard	11 passed; production build passed
Native Wi-Fi	10 passed
Total local tests	231 passed; no failures/skips in these suites
Android/iOS JS export	Passed; bundling is not hardware validation
Android arm64 APK	Native build passed; acoustic class present in DEX; signature verified
iOS simulator	Xcode BUILD SUCCEEDED; SpvsAcoustic Swift source compiled
iOS device app	Signed generic-device build passed; codesign verification passed
Native configuration	Microphone/module declarations verified
Browser E2E	Upstream cloud's 3 passing cases retained; not rerun against the local live demo database
Containers	Upstream build evidence retained; not rebuilt/deployed locally in this update

The cloud report's total is 234 because it includes three browser tests. The fresh local total is 231; do not add repeated runs as new independent cases. Upstream deprecation/SDK warnings remain. Actual local results and hashes are in results.json.

The rebuilt local Android APK is 78,138,028 bytes; SHA-256 `6cd0d397cccbbc42f6033c9a98eeab5a7c0d2f3f3d312b9dfdef72c607ab35c2`. The cloud-only APK had a different build hash and was not distributed in Git. The local replacement includes the required native microphone module.

Compatibility and policy transition

New attendance requires a rotating QR proof and a server-verified acoustic response before face/liveness, device, GPS and platform-presence checks can pass. Faculty must start challenges, display QR and play the queued student's signal. The capture is foreground/bounded and PCM stays transient on the phone.

QR rotates every five seconds with at most ten-second validity; audio redemption has a server-clock five-second deadline. Individual audio playback reservations serialize the channel to at most 12 per minute. Recordings/live relay and sound leakage remain possible; this is not distance-bounding or certified proximity.

The migration retires old iOS gps_only policy into dynamic if present, but the local review policy was unchanged. Dynamic iOS approval is a separate audited administrator choice and was not enabled here. Provider attestation remains unconfigured; QR/audio proof is not independent platform/sensor attestation.

Previously installed phone builds lack SpvsAcoustic and must be replaced before testing the new flow. **No rebuilt client was installed or launched on a phone during this update.** The signed iPhone app remains in ignored local build output; no signing keys/profile or production IPA is published. Native compilation is not microphone permission,

speaker/microphone or acoustic-channel validation.

Security, privacy and remaining work

Fresh npm audit: 4 high affected mobile entries and zero dashboard findings. No forced dependency downgrade was applied. Backend vulnerability scanning is historical, not a current clearance. Genuine provider attestation, calibrated biometrics, load/throughput, retention/backups, production TLS/signing and institutional policy remain release gates.

The C003 report records the earlier inside automatic pass and the corrected hostel GPS rejection. They predate policy v2 and do not demonstrate physical QR/audio success. The old immediate-doorway Wi-Fi-specific attendance test was not completed.

All physical testing waits for explicit user readiness in the classroom. Secrets, database backup/data, raw AP identifiers, exact locations, private keys, biometric templates, raw frames/audio and raw UI snapshots stay excluded from Git.